

5G-TSN/DetNet Integrated Networks for Industrial IoT:

An ML Framework to Balance Reliability and Bandwidth Utilization

Mohammed Abuibaid*, Amir Hoseein Ghorab*, Aysun Aslan Saruhan*, Marc St-Hilaire*, Glenn Parsonst†, János Farkast‡, Balázs Varga†, Vicknesan Ayadurai†, István Moldován‡, Miklós Máté‡

* Carleton University, Ottawa, ON, Canada

† Ericsson

‡ Budapest University of Technology and Economics, Budapest, Hungary

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Agenda

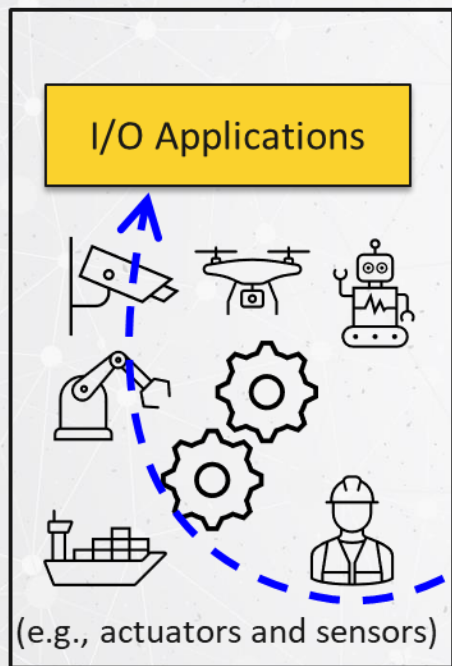
- Introduction
 - Industry 4.0, 5G uRLLC, TSN/DetNet, TSC
 - Future Industrial Environments
- Reliability Mechanisms in 5G-DetNet/TSN Integrated Networks
 - TSN and DetNet Reliability Functions, 5G Reliability Mechanisms
- Carleton 5G Testbed Architecture
 - 5G Network Setup, UE and MEC Hardware and Software Setup
 - Emulating Challenging Radio Conditions using Shielding Fabric
- Experiment Results and Discussion
- Conclusion and Outlook
- Q&A

Introduction

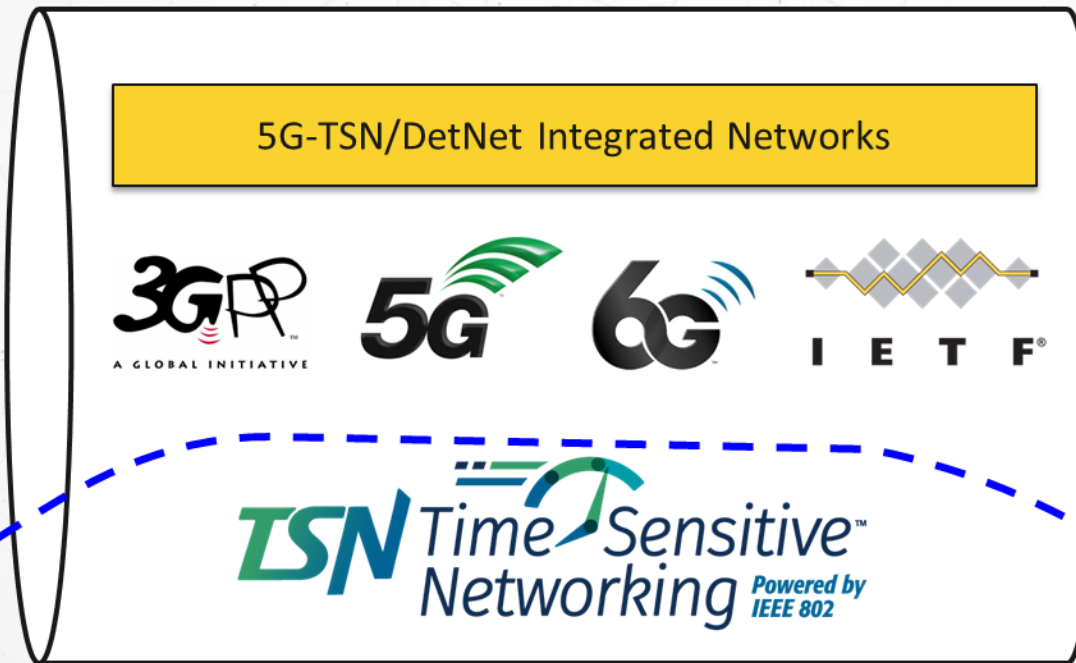
- Industry 4.0 aims for flexible, automated production with reduced costs.
 - Wireless communications enhance the adaptability/flexibility of industrial production lines.
 - To ensure safety and productivity, wireless solutions must meet strict timing and reliability requirements.
- Technical Insights on 5G-DetNet/TSN Integration:
 - 3GPP Release 16+ enhances support for TSC (Time-sensitive Communications) framework.
 - Data plane integration focuses on redundancy architectures, ensuring reliability against potential radio link or node failures.
- Real-world Implementation and Case Study:
 - Despite mature standardization, full implementation of 5G-DetNet/TSN in real-world industrial settings is still in progress.
 - This paper considers an aspect under discussion in 3GPP Rel19+ for balancing between network reliability and bandwidth utilization
 - The paper introduces a Machine Learning (ML)-based methodological framework to optimize reliability and bandwidth, paving the way for more resilient and adaptive communication for IIoT communications.

Future Industrial Environments

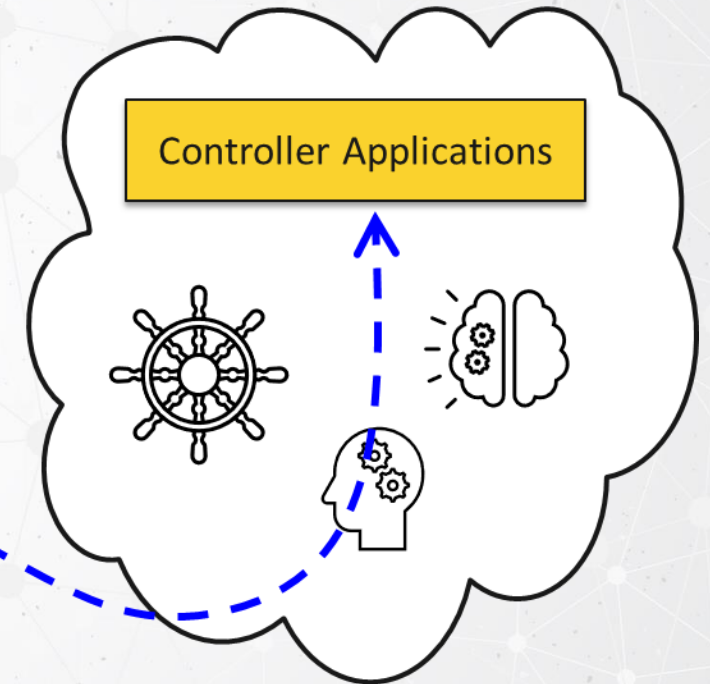
IIoT Devices



Industrial Communication Networks



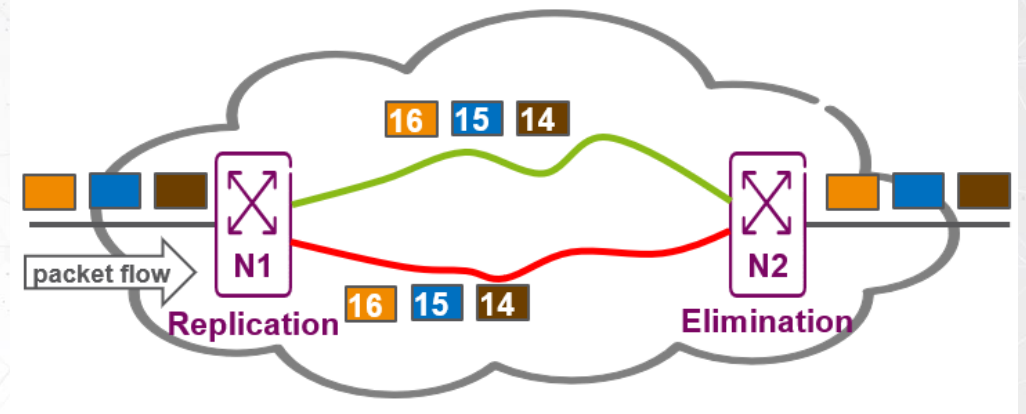
Edge Computing



TSN and DetNet Reliability Functions

- TSN and DetNet Standards for Reliability:

- IEEE 802.1CB and IETF RFC 8655 & RFC 8938.
- Mitigate packet loss from hardware faults or link errors w/o requiring failure detection or fallback mechanisms.

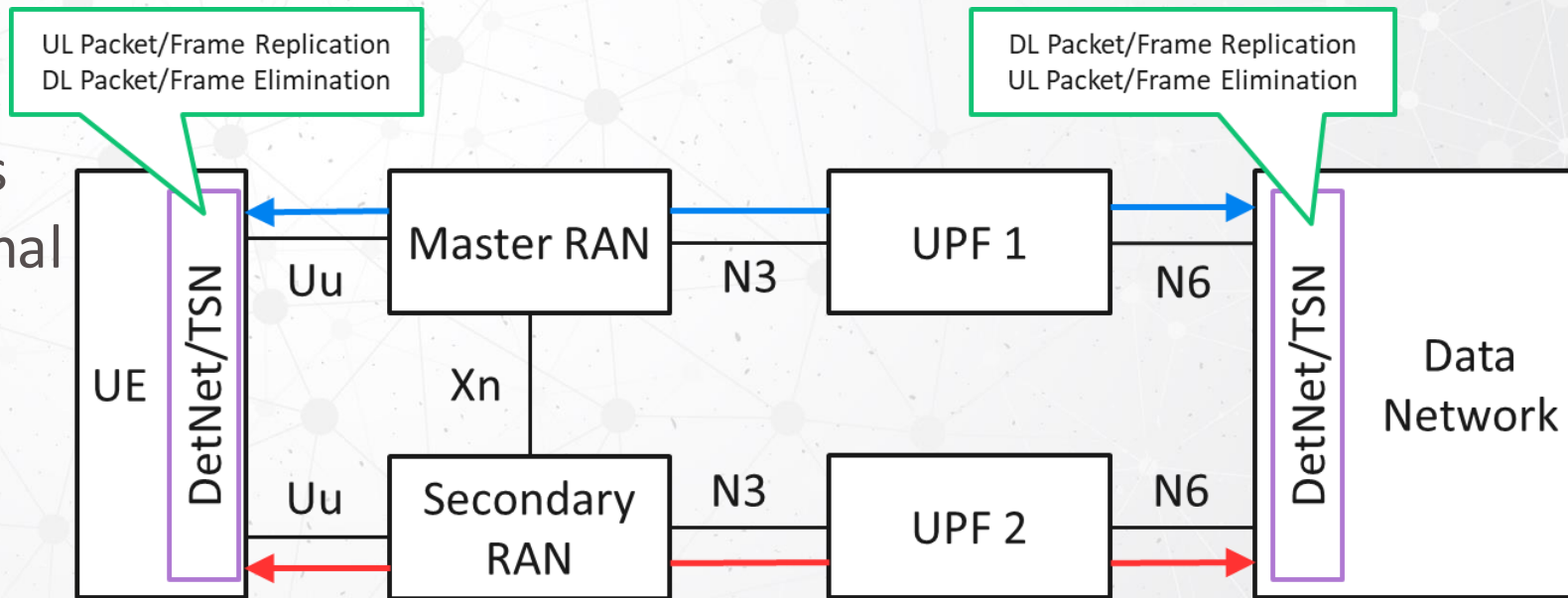


- Key Reliability Functions - Replication and Elimination:

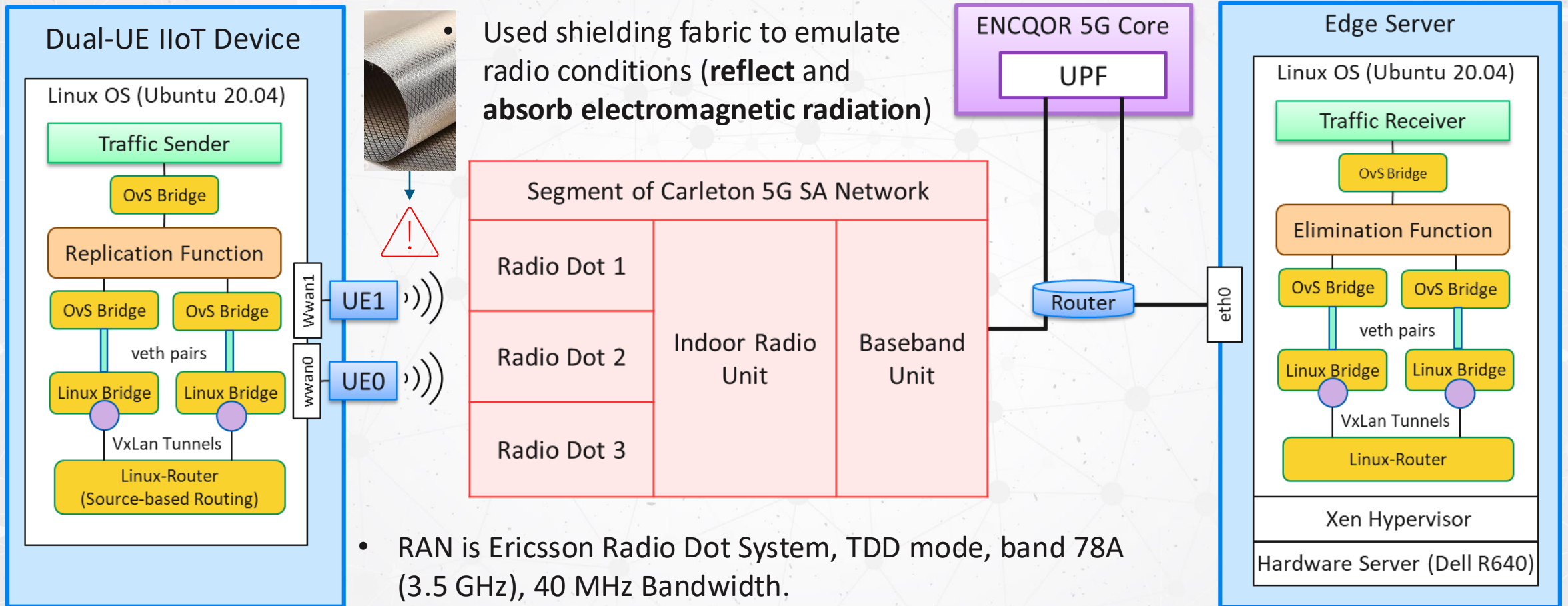
- Replication Function sends packet replicas through multiple disjoint paths to minimize identical packet loss.
- Elimination Function discards redundant packet replicas, ensuring only one copy proceeds towards the final application endpoint.
- Unique identifiers for TSN streams and DetNet flows, as well as individual packet tags, are essential for these operations, sourced from packet header fields and complemented by distinct metadata identifiers.

Redundancy Architecture for 5G-DetNet/TSN Integration

- The redundancy architecture establishes redundant user plane paths based on 5G Dual Connectivity technique.
 - Two independent PDU sessions are established targeting the same Data Network.
 - This ensures reliability against potential failures in the RAN and Core networks.
- DetNet/TSN reliability standards are implemented beyond/external to the 5G Network.



Carleton 5G Testbed Architecture

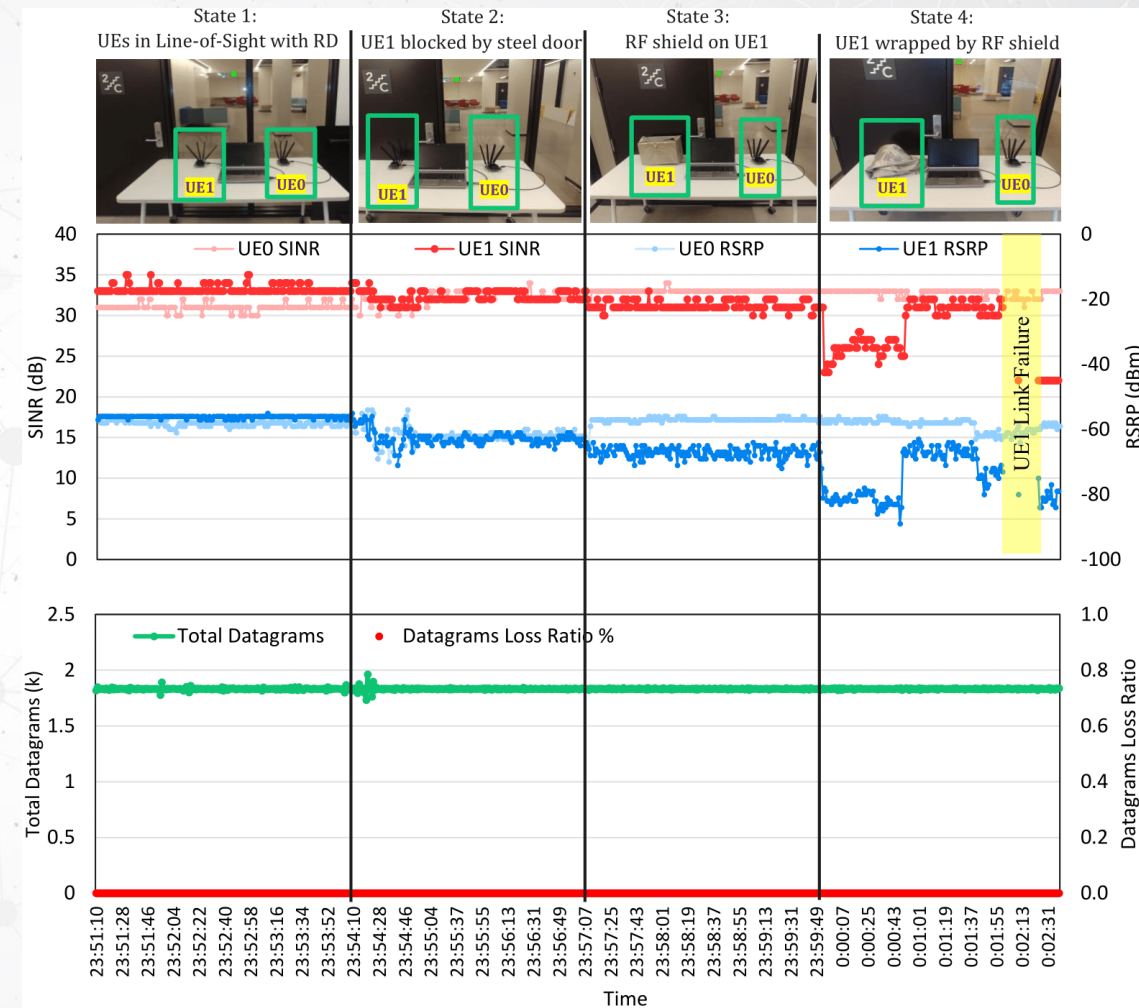


- Used shielding fabric to emulate radio conditions (**reflect** and **absorb electromagnetic radiation**)
- RAN is Ericsson Radio Dot System, TDD mode, band 78A (3.5 GHz), 40 MHz Bandwidth.
- UE is Quectel RM502Q-AE 5G module with PCIe EVB Kit.
- TSN/DetNet is Ericsson's Linux user-space implementation of IEEE FRER and IETF PREOF functions for R&D activities.

5G-TSN/DetNet Integration Experiments & Key Results:

Reliability Needs More Bandwidth Usage

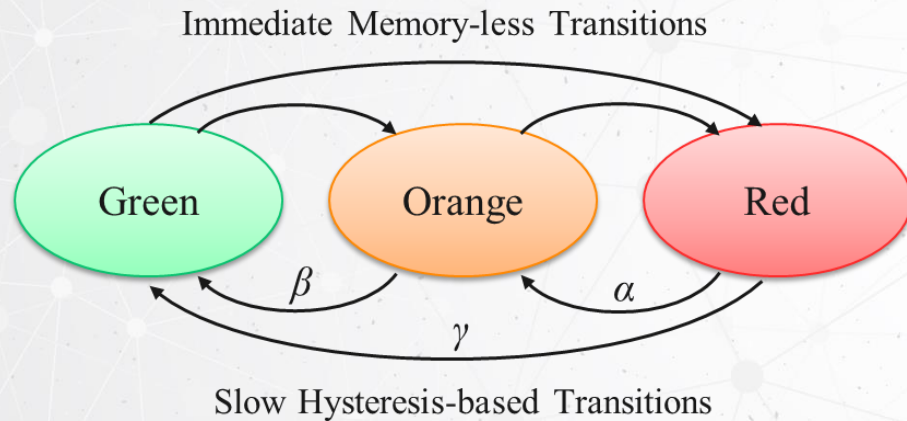
- **What?** Current 3GPP Rel16+ specifies an always-on fashion for TSN/DetNet replication functions.
- This is a redundancy overkill. **Why?**
 - Built-in 5G Reliability mechanisms: Tx/Rx Diversity, HARQ, PDCP Duplication + DC, Link Adaptation, mMIMO, etc.
 - TSN/DetNet standards allow for 1+n packet redundancy, but limited to 1+1 model in 3GPP standards. **Why?**
 - Hardware-based redundancy (cost and implementation practicality)
- **How?** An approach to control replication decisions.
 - Balancing the high reliability with bandwidth efficiency.
 - Reaching optimal network reliability while conserving bandwidth usage.



ML-based framework for TSN/DetNet Reliability Functions

Clustering Models:

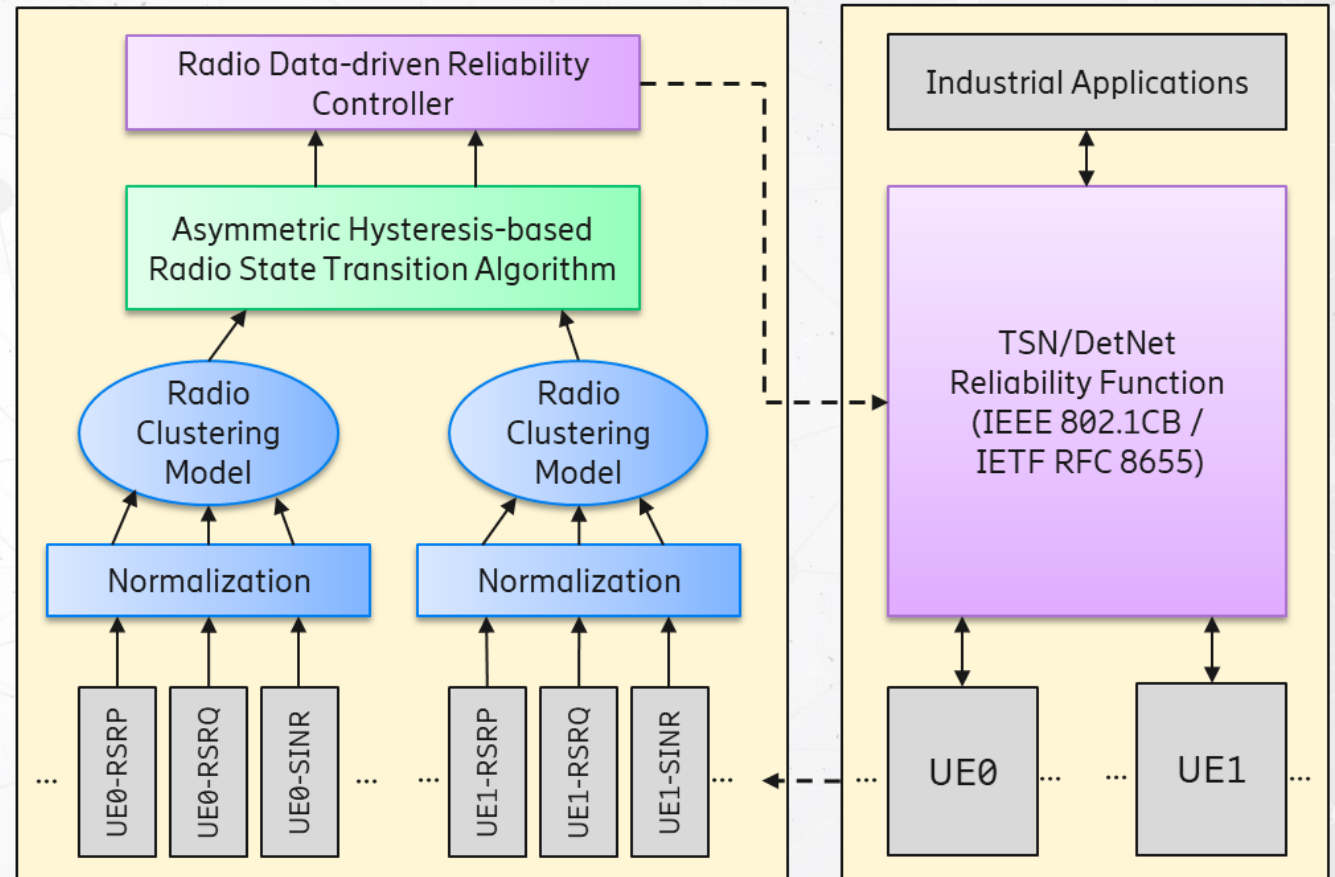
- Allow identifying patterns within radio conditions without requiring extensive dataset labeling.



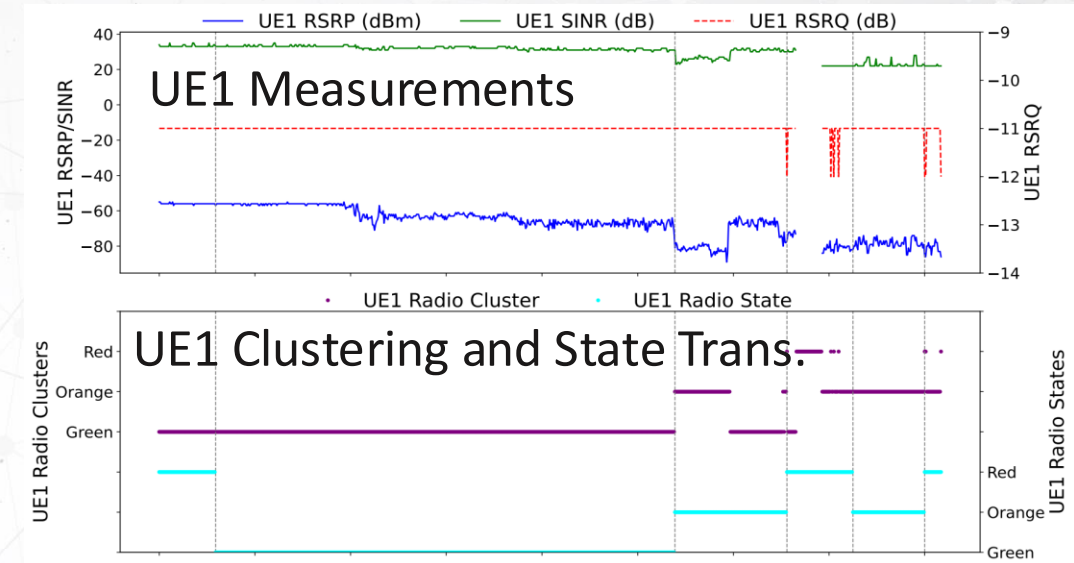
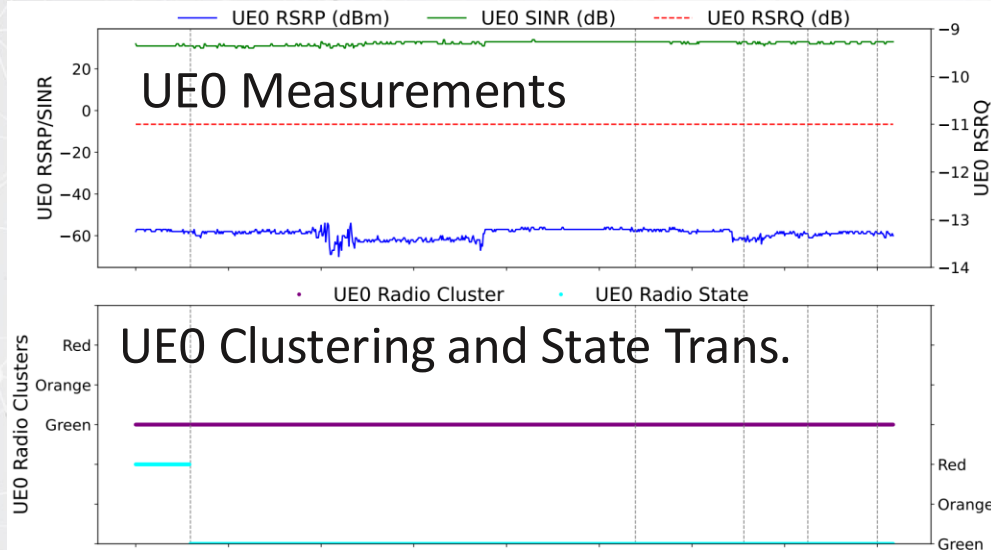
Asymmetric Hysteresis:

- Memory-less and memory-based k-means algorithm for state transitions

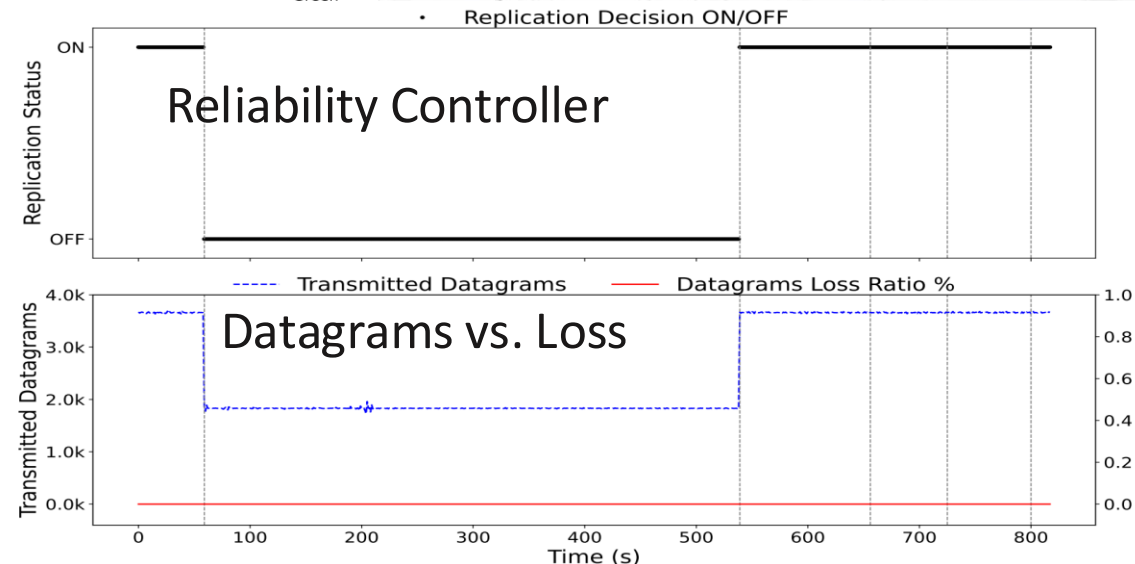
ML Framework for TSN/DetNet Reliability Functions



ML-based framework for TSN/DetNet Reliability Functions: Results and Discussion



Reduced total transmitted datagrams during favorable radio conditions



Maintained zero datagram loss during challenging radio conditions

Conclusion and Outlook

- **ML Framework for 5G-TSN/DetNet:**
 - Introduced a Machine Learning-based framework to dynamically optimize the trade-off between network reliability and bandwidth utilization in integrated 5G-TSN/DetNet networks for IIoT applications.
- **Real-Time Adaptive TSN/DetNet Network Management:**
 - Employed clustering and hysteresis logic to adaptively manage the activation of TSN/DetNet reliability features based on real-time radio conditions, demonstrating zero packet loss and efficient bandwidth utilization in a dual-UE 5G testbed.
- **Future Work & Broader Impact:**
 - Integrating the framework into real 3GPP systems, exploring alternative clustering algorithms, and expanding ML applications to automate network management tasks like resource allocation and fault recovery.

Thanks for Attention! Q&A



Mohammed Abuibaid 🍉
#CeaseFireNow



Amir Hoseein Ghorab



Aysun Aslan Saruhan
PhD Candidate at Carleton University

